# Adriel Ventura

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## EDUCATION

**Hofstra University** **Expected Dec. 2027**
*B.S. Mechanical Engineering, Aerospace Concentration* *Hempstead, NY*
Rabinowitz Honors College; Provost Scholar

## ENGINEERING EXPERIENCE

**Stellant Systems** **May 2026 – Present**
*Mechanical Engineering Co-op* *Melville, NY*

- Support mechanical design, vibration analysis, and thermal simulation of RF/microwave power-amplification hardware for aerospace defense programs.
- Perform steady-state thermal and mesh-refinement simulations in ANSYS Mechanical on PCB assemblies and heat pipes, evaluating boundary conditions up to 270°C while mitigating galvanic corrosion and environmental sealing risks.
- Programmed an automated macro to extract 3D CAD geometry data into Excel for BOM cost analysis, and processed Engineering Change Notices (ECNs) via SharePoint.
- Designed and manufactured a guillotine-style multi-diameter O-ring gasket cutter fixture, iterating through 4 prototypes via 3D printing and tolerance stack-up analysis.
- Converted legacy 2D AutoCAD geometry into 3D Autodesk Inventor models by translating sketch layers, reducing cycle time from 3 days to 1 day (67% faster).

## AEROSPACE & AUTOMATION PROJECTS

**CPU Heat Sink Thermal Design** / *MATLAB, Python, Fusion 360* **Spring 2026**

- Designed a 15-fin Al 6061 forced-convection heat sink for a 100 W load; selected two fewer fins than the thermal optimum to improve manufacturability while predicting approximately 70°C operation against an 85°C limit using MATLAB and Python scripts.

**NASA L'SPACE Mission Concept Academy** / *Thermal Engineer* **Fall 2026**

- Serve as Thermal Engineer for a multidisciplinary team designing a lunar exploration space-mission concept, managing thermal-subsystem parameters and payload viability.

## LEADERSHIP & RECOGNITION

- **President & Chapter Revamp Lead, Hofstra SHPE and AIAA** August 2026 – Present
  Reactivate two dormant engineering organizations by recruiting executive boards, rebuilding membership, coordinating recognition, and restoring technical programming.
- **President, Hofstra ASME** June 2026 – Present
  Lead executive-board operations and organized two technical networking events connecting 20+ students with industry mentors.
- **National Finalist (1 of 27), SHPE x Rockwell Automation LIVE AI Hackathon** June 2026 – Current
  Selected out of 54 competing collegiate teams to develop and pitch real-world industrial AI, machine learning, and autonomous automation solutions using Databricks and Python live at the 2026 National Convention.

## TECHNICAL SKILLS

**Programming & Software:** Databricks, Microsoft Azure, Unity, AI/ML Tools
**CAD & Modeling:** Autodesk Inventor, Siemens NX, SOLIDWORKS, Fusion 360, AutoCAD
**Analysis & Simulation:** ANSYS Mechanical (Thermal FEA, Vibration, Mesh Refinement)
**Manufacturing & Standards:** GD&T (ASME Y14.5), Tolerance Stack-Up Analysis, DFMA, Additive Manufacturing